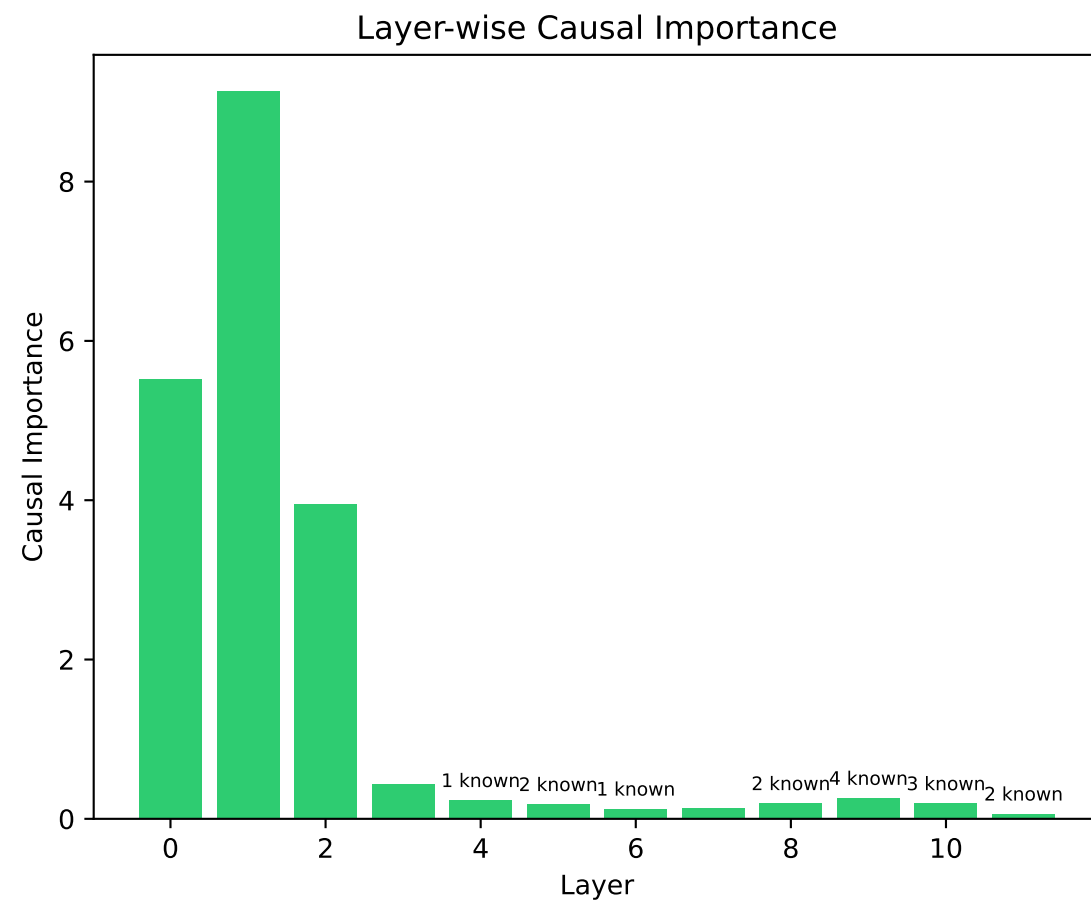
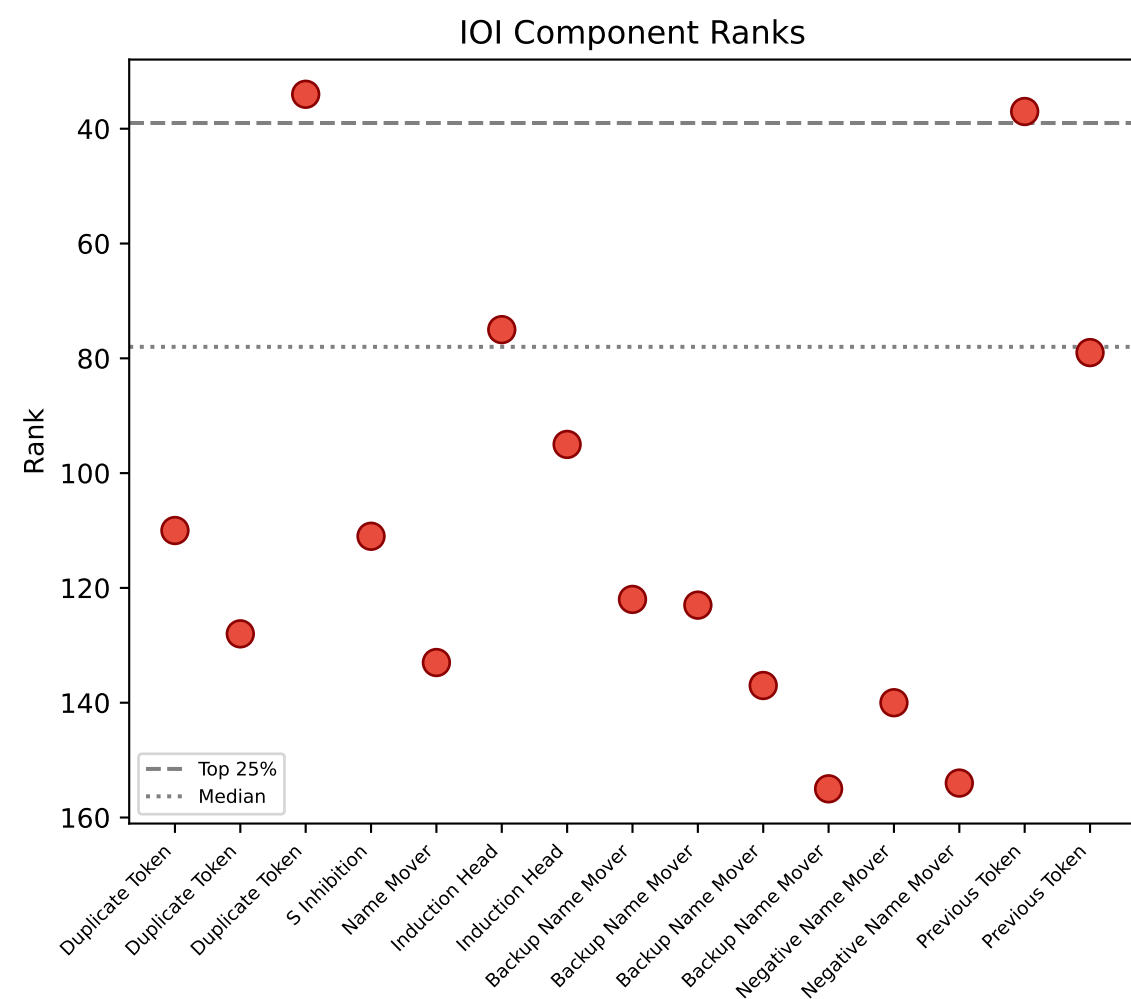
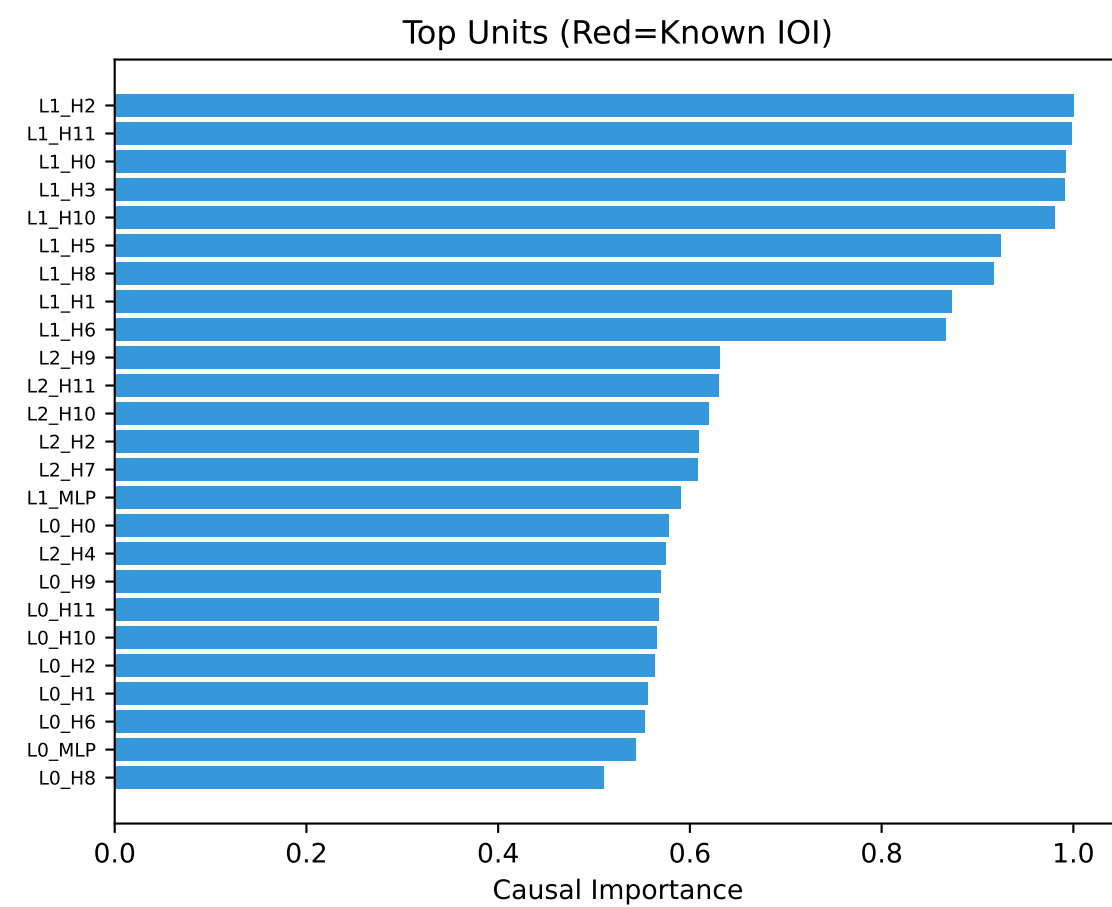
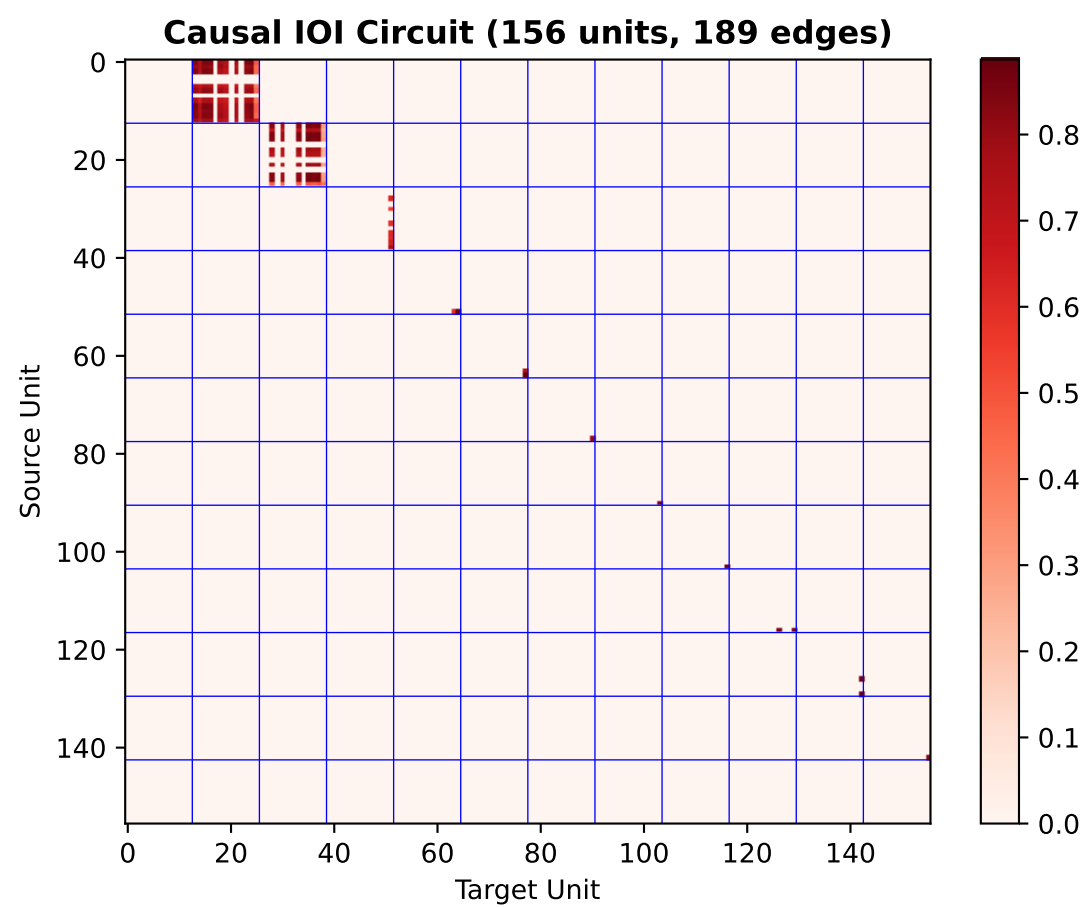




Causal IOI Circuit Extraction

Method: Activation Patching
Clean vs Corrupted IOI prompts
Causal signal = $|A_{\text{clean}} - A_{\text{corr}}|$

Results:
Units: 156 (12 layers $\times$ 13 units)
Edges: 189 ($\tau=0.15$)
Density: 0.8%

Recovery:
Components: 0/7
Known units: 15
Complexity: $O(1)$ vs ACDC $O(N^2)$
Queries: 2 forward passes